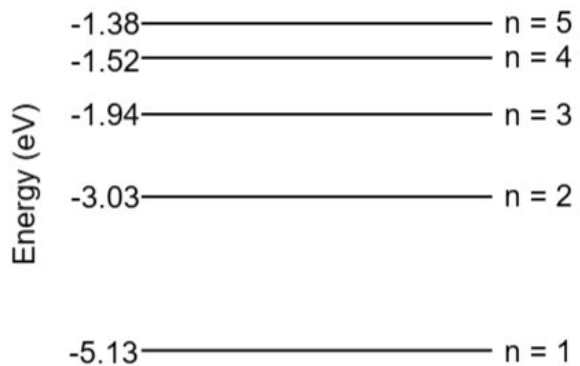
28 A gas has the following energy levels. It is excited by incoming electrons of kinetic energy 3.70 eV. Assume that the gas is cool and that all excitations of the gas by the incoming electrons are from the ground state.



The photons emitted on subsequent de-excitation are incident on a metal with work function energy 1.50 eV.

How many emitted photons of unique wavelength are capable of causing photoelectric effect on the metal?

A 3 **B** 4 **C** 6 **D** 10